Java Collection Framework

Pre-requisite topics for collection framework:-

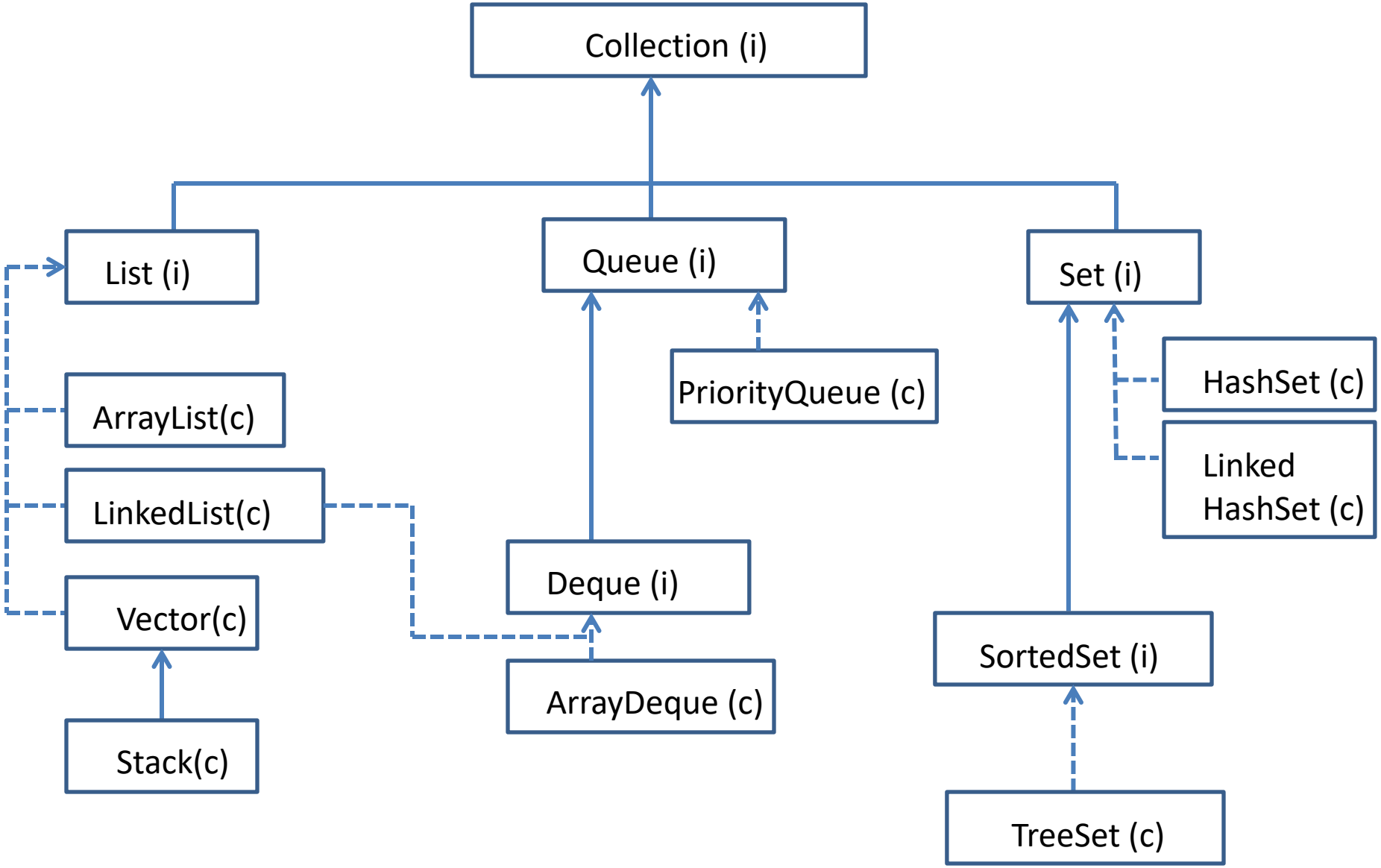
1. Type-casting
2. Interfaces
3. toString() method
4. for-each loop
5. Implementation classes
6. compareTo() method
7. Anonymous inner classes

Collection Framework:-

- ✓ Collection Framework contains group of classes and interfaces. By Using these classes and interfaces we are representing group of Objects as a single entity.
- ✓ Collection is sometimes called container. And it is object that group Multiple elements into a single entity.
- ✓ Collections are used to store, retrieve and manipulate data.

Note:- The root interface of Collection framework is **Collection.**

Hierarchy of Collection Framework



Java.util.ArrayList

ArrayList is extending AbstractList class and List interface. It is widely used class in Projects because it is providing functionality and flexibility.

ArrayList Characteristics:-

1. ArrayList is introduces in Java 1.2 Version.
2. ArrayList stores Heterogeneous objects.
3. Inside ArrayList we can store null objects.
4. ArrayList preserved iteration order it means whatever the order we inserted the data in Same way output is printed.
5. Duplicate Objects are allowed.
6. By using cursor we are able to retrieve the data from ArrayList: Iterator, ListIterator

Note:- There are two ways to iterate the elements in Java Collection:-
i.) By Iterator interface
ii.) By for-each loop

Java.util.LinkedList

Java LinkedList class uses doubly Linked list to store the elements. It provides a linked list Data Structure. It Inherits the AbstractList class and implements List and Deque interfaces.

- ✓ Java LinkedList class can contain duplicate elements.
- ✓ Java LinkedList class maintains iteration order.
- ✓ Java LinkedList class used as List, Stack or Queue.